FINTECH UTD · PROJECT PROPOSAL · BUILT ON JACK HENRY

FlowShift

Autopilot for small-business cash.

Know what’s safe to spend. Every morning. Automatically.

FlowShift forecasts a business’s next 14 days of cash, protects payroll, and moves every payment at the cheapest safe moment on the right rail — ACH, RTP or FedNow — from inside the bank app the owner already uses.

36.2M	$10M	1,000+
U.S. small businesses	RTP + FedNow per-payment limit	banks & CUs on Banno

Primary category: Transaction Processing / Money Movement · Secondary: Financial Health, Risk & Resilience

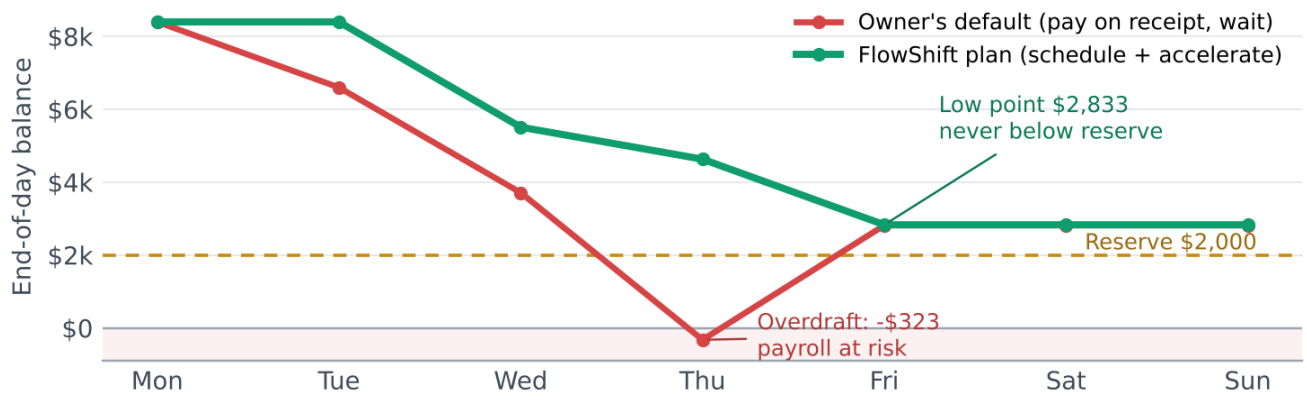
01 · THE PROBLEM

Small businesses don’t fail from lack of profit. They fail from bad timing.

Owners can see their balance. What no bank tells them is the thing they actually need: **what can I safely spend today, what should I pay when, and is paying for speed worth it?** Today they answer that in their head, on a spreadsheet, or at 11pm the night before payroll. A timing mistake turns a profitable week into an overdraft, a bounced payroll, or a high-interest advance.

A real week, simplified

Item	Amount	When	Notes
Checking balance (Monday)	$8,400	Now	Owner wants $2,000 kept as a reserve
Supplier invoice	−$1,800	Due Friday	Owner habit: pay on receipt (Tuesday)
Rent + insurance (auto-debit)	−$2,888	Wednesday	$2,600 rent, $288 insurance
Payroll	−$4,000	Thursday	Cannot be late
Card settlement	+$3,140	Friday (standard)	Instant payout available Thursday for 0.6% ($18.84)



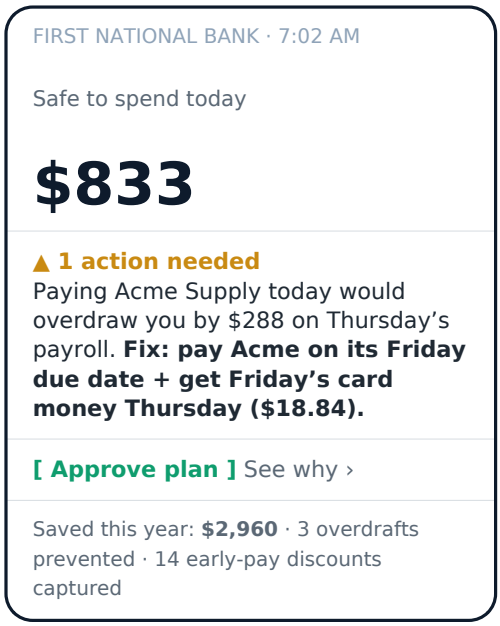
Plan	Thursday balance	Lowest point	Cost	Outcome
Owner’s default: pay supplier Tue, wait for settlement	-$323	-$323	$35 overdraft fee	**Payroll at risk**
Just reschedule supplier to Friday	$1,512	$1,512	$0	**$488 below reserve**
FlowShift: schedule supplier for Fri + accelerate settlement	$4,633	$2,833	$18.84	**Payroll safe, reserve intact, supplier paid on time**

The insight: no single move fixes this week. Rescheduling alone still breaches the reserve; paying for speed alone wastes money in most weeks. The right answer is a *combination*, recomputed every day. That is an optimization problem — and it is exactly the kind owners will never do by hand.

02 · THE PRODUCT

One number every morning. One tap to act.

Dashboards tell owners what already happened. FlowShift tells them what to do next — and, once trusted, does it for them. The whole experience is built around a single number owners will check like the weather: **Safe to Spend**.



Safe to Spend. Balance minus everything committed over the next 14 days minus the reserve, using a conservative forecast. One number, updated every time money moves.

Payroll Shield. Payroll is a hard constraint. If any forecast path puts it at risk, FlowShift alerts 72 hours ahead with a fix already built.

Smart Pay. Every bill gets a recommended pay date and rail. Pay on the due date, not on receipt; use instant rails only when they earn their fee.

Discount Hunter. Takes "2/10 net 30" early-payment discounts *only* when cash allows. A 2% discount for paying 20 days early is a ~37% annualized return.

Savings Receipt. A running total of fees avoided, discounts captured and overdrafts prevented. The product proves its own value every month.

Why owners keep coming back

Hook	What it feels like to the owner
Daily ritual	A 7am push with one number replaces the anxious balance-checking they already do several times a day.
Trust ladder	**Suggest → Approve with one tap → Autopilot within rules** ("auto-approve anything under $2,500 that keeps me above reserve"). Control is never taken, only delegated.
Visible money	The Savings Receipt makes the value concrete. Nobody cancels the tool that saved them $3,000.
Zero setup	It lives in their existing bank app. No new login, no linking accounts, no bookkeeping. Bills are detected from history; owners just confirm.

Core thesis: the product is a deterministic decision engine, not a chatbot. Language models only *explain* decisions in plain English; every number shown comes from auditable math the bank can review.

03 · THE ENGINE

Forecast → Optimize → Explain → Execute

Stage	How it works
1. Forecast	Detect recurring flows (payroll, rent, subscriptions, card settlements) from transaction history. Produce a daily 14-day balance forecast with a **P10–P90 band**. Safety decisions use the pessimistic P10 path, so uncertainty makes FlowShift more careful, never less.
2. Optimize	For each obligation choose an action: pay now, schedule for date *d*, accelerate an inflow, or (if offered) draw a short credit line. Pick the plan that minimizes total cost while satisfying hard constraints.
3. Explain	Turn the optimizer's structured output into one or two plain-English sentences with the projected balance impact. Template-first, LLM-polished, always traceable to the numbers.
4. Execute	MVP: human approval, payments simulated. Production: the bank initiates ACH / RTP / FedNow through Jack Henry payments rails, under owner-set rules.

The objective, stated precisely

$$\text{minimize } \Sigma \text{ rail fees} + \Sigma \text{ late penalties} + \text{credit interest} - \Sigma \text{ early-pay discounts} + \lambda \cdot \Sigma_t \max(0, R - B_t)$$

$$\text{subject to } B_t^{P10} \geq 0 \text{ for every day } t \text{ (never overdraft, even in the bad case)}$$
$$\text{payroll and other hard deadlines are met on time}$$
$$\text{each payment uses a rail available to that bank and payee}$$

R is the owner's reserve and λ sets how much they hate dipping below it. With a 14-day horizon and a few dozen obligations, the problem is small enough to solve exactly in milliseconds with a constraint solver (OR-Tools CP-SAT), which means every recommendation is optimal and reproducible — important when a bank has to defend it.

Rail logic in one table

Rail	Speed	Typical cost	FlowShift uses it when…
Standard ACH	1–2 business days	Lowest	The default. Schedule so funds land on the due date.
Same-day ACH	Same business day	Low	A bill is due tomorrow and nothing was scheduled in time.
RTP / FedNow	Seconds, 24/7	Higher, per payment	Speed avoids a bigger cost: overdraft, late fee, missed discount, or a reserve breach.
Instant payout of card sales	Same day	% of amount	An inflow arriving one day earlier protects payroll or the reserve (the example above).

What makes this technically serious

- Time-series forecasting with calibrated uncertainty, evaluated by backtest, not vibes.
- Constraint optimization with real economic trade-offs, not if/else rules.
- Event-driven backend: every new transaction re-plans the next 14 days.
- Explainability and audit trail designed for a regulated buyer from day one.

04 · WHY NOW, AND WHY JACK HENRY

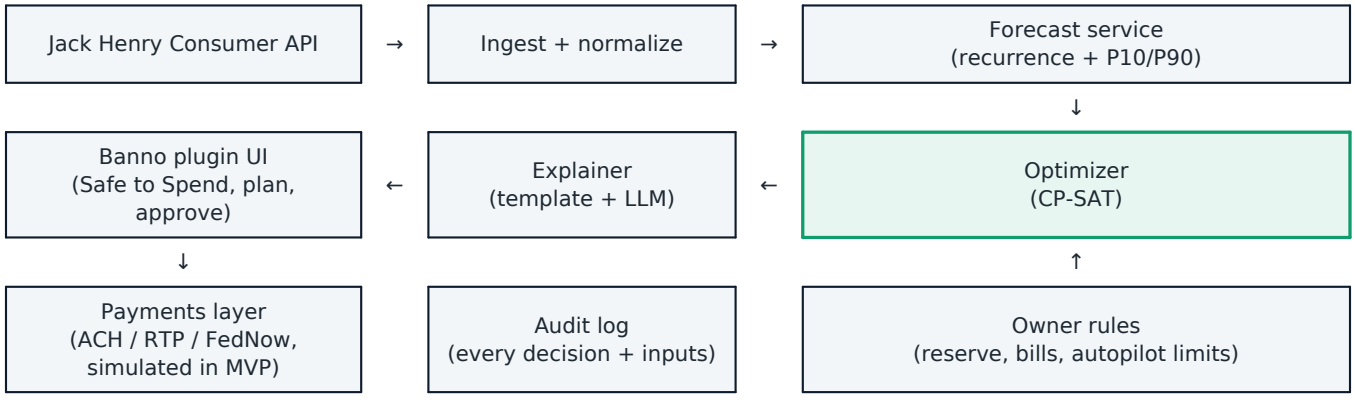
The rails just got fast. Nobody is deciding when to use them.

- **Instant payments went mainstream for business.** The RTP network raised its per-payment limit to $10M in February 2025, and FedNow raised its limit to $10M in November 2025, explicitly to support B2B and supplier payments. FedNow now has 1,500+ participating institutions.
- **Speed created a new decision problem.** When every payment could be instant, owners need something that knows when instant is worth paying for. Banks have the rails; they don’t have the brain that routes across them.
- **Community banks are losing small businesses to neobanks** that feel smarter. FlowShift lets a local bank ship a better cash experience than a neobank, without building a treasury product.
- **The distribution already exists.** Jack Henry reported 1,023 banks and credit unions on Banno (344 live with Banno Business), and 400+ clients each on RTP and FedNow, as of mid-2025. A Banno plugin reaches all of them through one integration.

Built on what Jack Henry already exposes

Jack Henry capability	How FlowShift uses it
Banno Plugin Framework	Embed Safe to Spend and the daily plan inside the bank’s own digital banking app.
Consumer API + OAuth / OpenID Connect	Read balances and transactions with the customer’s permission. No credential handling, no screen scraping.
Payments Orchestrator	Simulate, then execute, ACH and faster-payment workflows as the bank enables them.

Architecture



05 · THE BUSINESS

Everyone in the chain gets paid. That’s why it spreads.

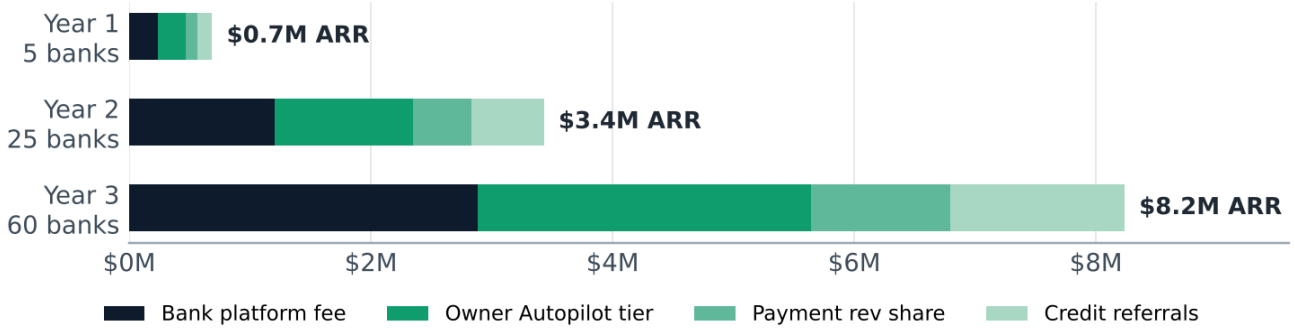
FlowShift is sold **B2B2B**: to banks and credit unions, who offer it to their business customers under their own brand. It works commercially because each party gets a clear, measurable win.

Who	What they get	What they pay
Small-business owner	Fewer fees, captured discounts, payroll that never bounces, 10 minutes a week instead of hours of worrying.	Free basic tier via the bank; **Autopilot** tier ~$24/month
Bank / credit union	Stickier deposits, more payment volume on its rails, early warning of credit need, and a reason to win businesses back from neobanks.	~$5 per active business per month
FlowShift	Recurring software revenue plus usage-based upside that grows with every payment and loan it routes.	—

What it’s worth to one owner (illustrative ~$1.5M-revenue business)

Value driver	Assumption	Per year
Early-pay discounts captured	$40k/mo supplier spend, 25% offers 2/10 net 30	$2,400
Overdraft and late fees avoided	One $35 fee avoided per month	$420
Unneeded rush fees avoided	Instant rails used only when they pay off	$300
Total value	vs. $288/yr for Autopilot	**$3,120 (~11x)**

Four revenue streams, one customer



Model assumptions: 800 active businesses per bank (a community bank with ~3,200 business accounts at 25% adoption). Per business per year: bank platform fee $60 + Autopilot (20% upgrade at $24/mo) $58 + payment revenue share $24 + working-capital referral share $30 = **$172**. Year 3 = 60 banks × 800 businesses × $172 ≈ **$8.2M ARR**, from under 6% of Banno’s existing client base. Illustrative targets to validate, not a forecast.

The hidden asset is the data. FlowShift knows, days in advance, which healthy businesses are about to be short on cash. That is the best lending signal a bank can have: offer a small line of credit *before* the owner goes looking for a costly merchant cash advance. Banks will pay for that pipeline, and it is the path from a $5 feature to a core product.

06 · COMPETITION AND RISKS

Where we win, and what could break us

Alternative	What it does well	Why FlowShift still wins
Bank cash-flow charts	Already in the app	Descriptive only. Shows the past, never recommends an action.
Accounting-software forecasts	Rich bookkeeping data	Can’t move money or choose a rail, and needs clean books most small businesses don’t keep.
Neobanks for SMBs	Modern UX	Require switching banks. FlowShift brings the experience to the bank the owner already trusts.
Corporate treasury systems	Powerful optimization	Priced and built for CFOs with treasury teams, not a restaurant owner.
A spreadsheet + the owner	Free	Doesn’t run at 7am every day, doesn’t know rail costs, and forgets the insurance debit.

Risks we are designing around

Risk	Mitigation
Forecast is wrong	Decide on the pessimistic P10 path, show confidence, keep humans approving until backtests earn trust.
Liability for a bad payment	FlowShift recommends; the bank executes within owner-set rules, with a full audit log of inputs and reasoning.
Bank sales cycles are slow	Go through the Banno plugin marketplace and Jack Henry’s channel, start with a free pilot for owners to prove savings.
Data access and compliance	Permissioned OAuth through Jack Henry only. Production execution needs institutional approval, security review and compliance controls.

How we’ll know it works

Metric	MVP target
Forecast accuracy	≤ 10% error on 7-day balance in backtests on synthetic + sandbox data
Shortfalls prevented	Catch ≥ 90% of injected overdraft scenarios at least 48 hours ahead
Dollars saved	Median simulated savings per business per year, versus an “owner default” baseline
Would they use it?	Interview 15 SMB owners and 3 community-bank staff by week 4. Goal: 10 owners say they’d check Safe to Spend daily

Sources. The Clearing House, “Higher $10 million RTP network transaction limit” (Dec 2024; effective Feb 9, 2025). Federal Reserve Financial Services, FedNow network limit increase to $10M (effective Nov 12, 2025) and 2026 fees announcement (1,500+ participating institutions). Jack Henry FY2025 Q4 earnings call (Banno: 1,023 clients, 344 live with Banno Business; 414 RTP and 401 FedNow clients). SBA Office of Advocacy, Frequently Asked Questions About Small Business 2026 (36.2M small businesses). Platform documentation: jackhenry.dev. Example week, savings and revenue figures are illustrative.

07 · THE SEMESTER PLAN

Small enough to ship. Sharp enough to win the room.

MVP component	Definition of done
1. Bank data	Jack Henry sandbox user connected; balances and transactions ingested and normalized.
2. Recurring detection	Payroll, rent and settlements detected automatically; owner confirms with one tap.
3. 14-day forecast	Daily balance forecast with P10–P90 band and reserve warnings.
4. Optimizer	CP-SAT plan choosing pay now / schedule / accelerate per obligation; reproduces the example week exactly.
5. Owner experience	Safe to Spend, Savings Receipt, plain-English reasons and one-tap approval in a Banno-plugin-style UI, with an audit log behind every action.

Team

Backend / integrations
Jack Henry APIs, OAuth, ingestion, payments simulator, audit log

Forecasting / optimization
Recurrence detection, probabilistic forecast, CP-SAT, backtests

Frontend / product
Banno plugin UI, Safe to Spend, approvals, Savings Receipt

Explainability + research
Explanation layer, interviews, 200-business demo dataset

Timeline

Weeks	Milestone
1–3	Jack Henry sandbox, data model, ingestion. Start owner and bank interviews.
4–6	Recurrence detection + forecast with uncertainty bands; backtest harness.
7–9	Optimizer, reserve logic, rail cost model; reproduce the example week.
10–11	Banno UI, Safe to Spend, explanations, Savings Receipt.
12–13	Edge cases, 200-business simulation, savings numbers for the pitch.
Demo day	Live: a business heads toward a bounced payroll; FlowShift sees it 3 days out, builds the $18.84 fix, owner taps approve.

Why join this project

- Real Jack Henry integration, not a mock fintech wrapper.
- Real engineering depth: forecasting, optimization, event-driven systems, explainable AI.
- A demo anyone gets in 30 seconds, a business model a bank would pay for, and a credible path to a pilot or startup.

The pitch in one line: FlowShift is autopilot for small-business cash — it tells owners what’s safe to spend every morning, moves every payment at the cheapest safe moment, and turns the bank they already use into the smartest one they’ve ever had.